

Proof of Conjecture 3: The Period Length at $\omega = 1$

Dirk Kunert *

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Statement

Theorem 1. *Let $\alpha, \beta \in \mathbb{N}$ with $\gcd(\alpha, \beta) = 1$, and let $\omega = 1$. Then the period length of the distance sequence $(\delta^{(i)})$ of the cut-and-project sequence is*

$$\lambda_{\alpha, \beta} = \Lambda_{\alpha, \beta} := \alpha + \beta + 1.$$

Proof

Step 1: The window constraint as a single invariant

For $\omega = 1$ the valid lattice points $(x, y) \in \mathbb{Z}^2$ are those satisfying

$$\frac{\alpha}{\beta}(x - 1) \leq y \leq \frac{\alpha}{\beta}x + 1.$$

Multiplying through by $\beta > 0$ gives the equivalent condition

$$\alpha x - \alpha \leq \beta y \leq \alpha x + \beta,$$

which can be rewritten as

$$-\beta \leq \underbrace{\alpha x - \beta y}_{=: r} \leq \alpha.$$

The invariant $r := \alpha x - \beta y$ therefore takes values in the integer set

$$\mathcal{R} := \{-\beta, -\beta + 1, \dots, 0, \dots, \alpha\}, \quad |\mathcal{R}| = \alpha + \beta + 1.$$

Step 2: Each value of r yields one arithmetic progression

Recall that the projected quantity used to compute distances is (see equation (7) of the companion paper [1])

$$\tilde{p} = \beta x + \alpha y.$$

For a fixed $r \in \mathcal{R}$, consider the linear system

$$\beta x + \alpha y = \tilde{p}, \tag{1}$$

$$\alpha x - \beta y = r. \tag{2}$$

The coefficient matrix has determinant $-\beta^2 - \alpha^2 = -(\alpha^2 + \beta^2)$, so the unique rational solution is

$$x = \frac{\beta \tilde{p} + \alpha r}{\alpha^2 + \beta^2}, \quad y = \frac{\alpha \tilde{p} - \beta r}{\alpha^2 + \beta^2}.$$

*Independent Researcher, Email: dirk.kunert@gmail.com

For $x, y \in \mathbb{Z}$ we need $(\alpha^2 + \beta^2) \mid (\beta\tilde{p} + \alpha r)$, i.e.

$$\tilde{p} \equiv -\alpha\beta^{-1}r \pmod{\alpha^2 + \beta^2},$$

where β^{-1} denotes the modular inverse of β modulo $\alpha^2 + \beta^2$ (existence is shown in Step 3). Denote this residue class by c_r . Then the set of all $\tilde{p}$ values compatible with invariant r is the arithmetic progression

$$P_r := \{ c_r + k(\alpha^2 + \beta^2) : k \in \mathbb{Z} \}.$$

Conversely, for every $k \in \mathbb{Z}$ the value $\tilde{p} = c_r + k(\alpha^2 + \beta^2)$ yields integer coordinates

$$x = \frac{\beta c_r + \alpha r}{\alpha^2 + \beta^2} + \beta k, \quad y = \frac{\alpha c_r - \beta r}{\alpha^2 + \beta^2} + \alpha k,$$

both of which are integers by the definition of c_r . Hence $P_r \subseteq S$ and

$$S = \bigcup_{r \in \mathcal{R}} P_r.$$

Step 3: The progressions are pairwise disjoint (Bézout)

Lemma 1. *The residues c_r are pairwise distinct modulo $\alpha^2 + \beta^2$.*

Proof. Suppose $c_{r_1} \equiv c_{r_2} \pmod{\alpha^2 + \beta^2}$ for $r_1, r_2 \in \mathcal{R}$. Then $(\alpha^2 + \beta^2) \mid \alpha(r_1 - r_2)$.

By **Bézout's identity**: $\gcd(\alpha, \beta) = 1$ implies $\gcd(\alpha, \beta^k) = 1$ for all $k \geq 1$ (any prime dividing both α and β^k would divide β , contradicting $\gcd(\alpha, \beta) = 1$). Therefore

$$\gcd(\alpha, \alpha^2 + \beta^2) = \gcd(\alpha, \beta^2) = 1.$$

Consequently $(\alpha^2 + \beta^2) \mid (r_1 - r_2)$. Since $|r_1 - r_2| \leq \alpha + \beta < \alpha^2 + \beta^2$ (for $(\alpha, \beta) \neq (1, 1)$, see Remark 1), we conclude $r_1 = r_2$. $\square$

The same argument gives $\gcd(\beta, \alpha^2 + \beta^2) = \gcd(\beta, \alpha^2) = 1$, so the modular inverse β^{-1} required in Step 2 exists.

Step 4: The sorted union has period $\alpha + \beta + 1$

Sort the $\alpha + \beta + 1$ distinct residues in ascending order:

$$0 \leq c_{(1)} < c_{(2)} < \dots < c_{(\alpha+\beta+1)} < \alpha^2 + \beta^2.$$

The sorted union S then reads

$$\dots, c_{(1)}, c_{(2)}, \dots, c_{(\alpha+\beta+1)}, c_{(1)} + D, c_{(2)} + D, \dots, c_{(\alpha+\beta+1)} + D, c_{(1)} + 2D, \dots$$

where $D := \alpha^2 + \beta^2$. The differences $\delta^{(i)} = \tilde{p}^{(i+1)} - \tilde{p}^{(i)}$ form the pattern

$$\underbrace{c_{(2)} - c_{(1)}, c_{(3)} - c_{(2)}, \dots, c_{(\alpha+\beta+1)} - c_{(\alpha+\beta)}, D - c_{(\alpha+\beta+1)} + c_{(1)}}_{\alpha+\beta+1 \text{ terms, summing to } D=\alpha^2+\beta^2}$$

and this pattern repeats exactly with period $\alpha + \beta + 1$. No smaller period is possible: the $\alpha + \beta + 1$ differences are determined by the $\alpha + \beta + 1$ distinct residues, and any sub-period would require a repeated residue, contradicting Lemma 1.

Therefore $\lambda_{\alpha, \beta} = \alpha + \beta + 1 = \Lambda_{\alpha, \beta}$. $\square$

Remarks

Remark 1. The case $\alpha = \beta = 1$ is degenerate: $\alpha^2 + \beta^2 = 2 < \alpha + \beta + 1 = 3$, so three distinct residues modulo 2 cannot all differ, and the progressions P_r overlap. In this case the sequence $(\tilde{p})$ contains duplicate values (e.g. both $(x, y) = (0, 1)$ and $(1, 0)$ project to $\tilde{p} = 1$), producing zero differences. Nevertheless, one can verify directly that the difference sequence is $(1, 0, 1, 1, 0, 1, \dots)$, which has period $3 = \Lambda_{1,1}$. The theorem therefore holds for $\alpha = \beta = 1$ as well, though the proof above does not cover this case.

Remark 2. The sum of one full period of differences equals $\alpha^2 + \beta^2 = D$, the common difference of the underlying arithmetic progressions. This can be used to verify numerical results: for $(\alpha, \beta) = (2, 1)$ the gaps $(1, 1, 1, 2)$ sum to $5 = 4 + 1$; for $(\alpha, \beta) = (3, 2)$ the gaps $(2, 1, 2, 3, 2, 3)$ sum to $13 = 9 + 4$.

References

- [1] Dirk Kunert, 2025, *Conjectures on the Period Lengths of One-Dimensional Cut-and-Project Sequences*, available at: <https://github.com/dkunert/cut-and-project>.